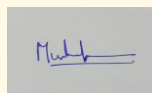


Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	SYED MUHAFIZ A	Reg. No.:	192472282
Section / Batch:		Date:	

Code of Conduct

I SYED MUHAFIZ A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:


Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

1. Morphological Analysis Pipeline

Input: *connected, connecting, connection*

Step 1: Rule-Based Decomposition

Word	Root	Suffix Type
connected	connect -ed	Inflectional (past tense)
connecting	connect -ing	Inflectional (present participle)
connection	connect -ion	Derivational (noun formation)

Step 2: Inflectional vs. Derivational

- **Inflectional:** connected, connecting (only grammatical changes)
- **Derivational:** connection (changes verb to noun)

Step 3: Normalization

All words are normalized to the common base form:

connect

Parsed Structure and Output

Word	Parsed Structure	Normalized Output
connected	connect + ed	connect
connecting	connect + ing	connect
connection	connect + ion	connect

2. Morphological Parsing Module

Input: *unhappy, happiness, happily*

Rule-Based Decomposition

Word	Prefix	Base Form	Suffix	Classification
unhappy	un-	happy	—	Derivational
happiness	—	happy	-ness	Derivational
happily	—	happy	-ly	Derivational

Semantic Consistency

- **un-** → Negates the meaning.
 - **-ness** → Converts adjective to noun.
 - **-ly** → Converts adjective to adverb.
-

All retain the semantic concept **happy**.

Morphological Breakdown

Word	Breakdown	Root
unhappy	un + happy	happy
happiness	happy + ness	happy
happily	happy + ly	happy

3. Stemming-Based Preprocessing

Input: *played, player, playing*

Morphological Rules

Word	Rule Applied	Stem	Classification
played	Remove -ed	play	Inflectional
player	Remove -er	play	Derivational
playing	Remove -ing	play	Inflectional

Root Extraction

All words are reduced to:

play

Final Output

Word	Stem	Type
played	play	Inflectional
player	play	Derivational
playing	play	Inflectional

4. Finite-State Morphological Parsing

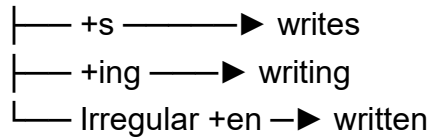
Input: *writes, writing, written*

State Transitions

Start



Root: write



Morphological Breakdown

Word	Breakdown	Pattern	Type
writes	write + s	Regular	Inflectional
writing	write + ing	Regular	Inflectional
written	write + en	Irregular	Inflectional

Normalized Output

All forms map to the common root:

write

Word	State Transition	Normalized Output
writes	write → +s	write
writing	write → +ing	write
written	write → +en (irregular)	write

5. Porter Stemmer-Based Preprocessing

Input: relational, relation, relate

Step-by-Step Stemming

Word	Stemming Step	Intermediate Form	Final Stem
relational	Remove -ational → -ate	relate	relat
relation	Remove -ion	relat	relat
relate	Remove final -e	relat	relat

Explanation

- relational
 - relational → relate (remove **-ational**)
 - relate → relat (remove final **e**)
- relation

- relation → relat (remove **-ion**)
-

- **relate**

- relate → relat (remove final **e**)
-

Final Output

Word	Intermediate Form(s)	Final Stem
relational	relate	relat
relation	relat	relat
relate	relat	relat

Uniform stem produced by the Porter Stemmer: relat

Rubrics for Calculation

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____